



IONQ INC. [IONQ]

2015-09-29, Innovator

"The Futer is Quantum"

Analysis Report

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1. Introduction to Core Business Segments and Roadmap

A U.S. quantum technology company developing an integrated quantum platform based on trapped-ion technology, spanning quantum computing, networking, cybersecurity, and sensing.

[1] Integrated Quantum Technology Platform

1) Quantum Computing

* Principle: Trapped-Ion Technology: Electrically charged atoms, or ions, are confined using electromagnetic fields, and their quantum states are controlled with lasers. Individual atoms or ions serve as the qubit medium, laser-based control is used, and qubits within the same ion chain have high connectivity.

* Revenue is generated through cloud-based quantum computing services, the sale and installation of dedicated quantum computers, and application development and technical service offerings.

2) Quantum Networking

* Principle: A technology for transmitting quantum information or entanglement between physically separated quantum devices.

* It presents a structured, systems-level approach encompassing quantum repeaters, networking equipment, quantum memory, and frequency-conversion technologies.

3) Quantum Cybersecurity

* Principle: A business focused on enhancing the security of communications and data.

* It develops systems for securely exchanging encryption keys using quantum-mechanical principles, including Quantum Key Distribution (QKD), Quantum Random Number Generators (QRNGs), and cryptographic key management systems.

4) Quantum Sensing

* Principle: A technology that uses the sensitivity of quantum states to precisely measure time, gravity, acceleration, magnetic fields, and other physical quantities.

* Key applications include Positioning, Navigation, and Timing (PNT) technologies such as atomic clocks, inertial sensors, and navigation systems.

Trapped-ion technology offers advantages in achieving high gate fidelity, long coherence times, and strong qubit connectivity. However, compared with other approaches, it has **relatively slower** computational speeds and faces greater **challenges** in large-scale expansion, equipment configuration, and maintenance.

[2] Roadmap

Date	System Stage	Key Performance Metrics	Achievement	Technical Significance
2021	Harmony	#AQ 9~11	Achieved	-
2022	Aria	#AQ 25	Achieved	-
2023	Forte	#AQ 29	Achieved	-
2024	Forte Enterprise	#AQ 36	Achieved	-
2025	Tempo	#AQ 64	Achieved	-
2025	Two-Qubit Gate Fidelity	99.99%	Achieved	-
2026	Chip-Based System	100~256+ Qbit	-	12 Logical Qubits
2027	Large-Scale Single QPU	10,000 Qbit	-	800 Logical Qubits
2028	Modular Interconnected System	200,000-Physical-Qubit Scale Test	-	8,000 Logical Qubits Test
2029	L-S Quantum Error Correction	200,000-Physical-Qubit	-	8,000 Logical Qubits
2030	Completion of a L-S System	2,000,000-Physical-Qubit	-	4~80,000 Logical Qubits

In the January 2026 announcement of the SkyWater acquisition, the company stated that it plans to begin functional testing of a **200,000-physical-qubit** QPU and **8,000 logical qubits** in **2028**. This can be interpreted as an acceleration of the testing timeline for a 200,000-qubit-class chip compared with the previously disclosed roadmap.

2. M&A History

Date	Company Name	Cost(unit : M\$)	Field
2023.01	Entangled Networks asset	-	Distributed Quantum Computing, Quantum Compilers
2025.01	Qubitekk asset	-	Quantum Networking, Entanglement Distribution
2025.04	ID Quantique Controlling Interest	116.2	QKD, QRNG, Single-Photon Detectors
2025.05	Lightsynq	306.2	Photonic Interconnects, Quantum Memory
2025.07	Capella Space	424.8	SAR Satellites, Space Infrastructure
2025.09	Oxford Ionics	1590	Chip-Based Ion Traps, Electronic Qubit Control
2025.10	Vector Atomic	181.5	Atomic Clocks, Inertial Sensors, Quantum Sensing
2026.01	Skyloom Global	190	Free-Space Optical Communications, Optical Terminals
2026.01	Seed Innovations	30.1	AI and Machine Learning, Cloud, Software

[M&A Table]

Initially, the company focused on developing trapped-ion quantum computers and providing cloud services. It later expanded the scope of its quantum technology applications by acquiring companies in networking, cybersecurity, sensing, and satellite communications. More recently, through the acquisitions of Oxford Ionics and SkyWater, it has sought to internalize quantum chip design, control, and manufacturing, while pursuing vertical integration across quantum computer development, communications, applications, and production.

2-1. Acquisition Purpose

1) 2025.05 **Lightsynq** acquisition

: Secured photonic interconnect and quantum memory technologies to connect multiple QPUs and accelerate the development of fault-tolerant quantum computers and the quantum internet.

2) 2025.07 **Capella Space** acquisition

: Aims to secure satellite infrastructure and establish a global space-based quantum key distribution and secure communications network.

3) 2025.09 **Oxford Ionics** acquisition

: Combines semiconductor chip-based ion traps with electronic qubit control technology to improve the scalability and manufacturability of large-scale, high-fidelity quantum computers.

3. Revenue and Profit Structure

3-1. Quantum Hardware Sales

Quantum computers such as Forte Enterprise and Tempo, along with quantum cybersecurity, networking, and sensing equipment, are supplied to government agencies, research institutes, universities, and large corporations. Large contracts account for a significant share of revenue, resulting in substantial revenue per contract but considerable quarterly volatility depending on manufacturing and installation schedules.

3-2. Cloud Quantum Computing

: Quantum computing access is provided through Amazon Braket, Microsoft Azure Quantum, Google Cloud, and the company's proprietary platform. Customers pay either a fixed subscription fee or based on actual usage, creating a structure with the potential to expand recurring revenue over the long term.

3-3. Dedicated System Access

A contract under which a government agency or company secures a specified capacity and usage time on a particular quantum computer for an extended period. Compared with general cloud access, these contracts are larger in scale and more likely to develop into long-term agreements.

3-4. Algorithm and Consulting Services

Provides co-development services that implement quantum algorithms for applications in drug discovery, materials science, finance, logistics, and defense. Even with the current performance limitations of quantum computers, the company is able to generate revenue through proof-of-concept projects and research and development services.

3-5. Maintenance, Training, and Technical Support

Provides maintenance, calibration, software updates, and user training following the installation of quantum computers and networking equipment. As the installed base of systems expands, this business generates an increasing amount of recurring revenue.

3-6. Quantum Cybersecurity and Networking Products

Uses technologies acquired through its subsidiaries to sell quantum key distribution equipment, quantum random number generators, single-photon detectors, and networking equipment. Its primary customers include telecommunications companies, government agencies, and research networks.

3-7. Quantum Sensing Business

Supplies atomic clocks, inertial sensors, and GPS-independent navigation equipment to defense, aerospace, and research institutions. This business expands revenue sources beyond quantum computing by addressing demand for precision navigation and measurement.

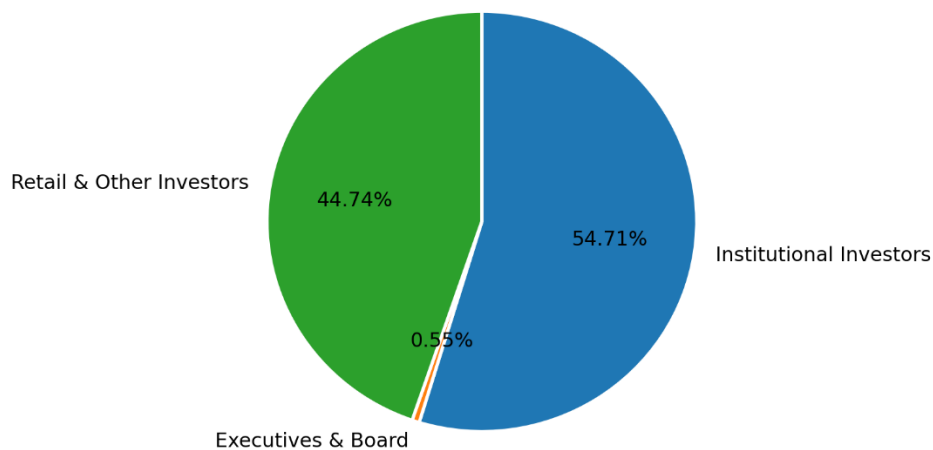
3-8. Satellite and Data Services

Provides satellite imagery, Earth observation data, and optical communications services through Capella Space and Skyloom. In addition to satellite hardware sales, it generates recurring revenue through data access fees and usage-based charges.

4. Ownership Structure and Financial Health

4-1. Investor Ratio

IonQ Shareholder Composition



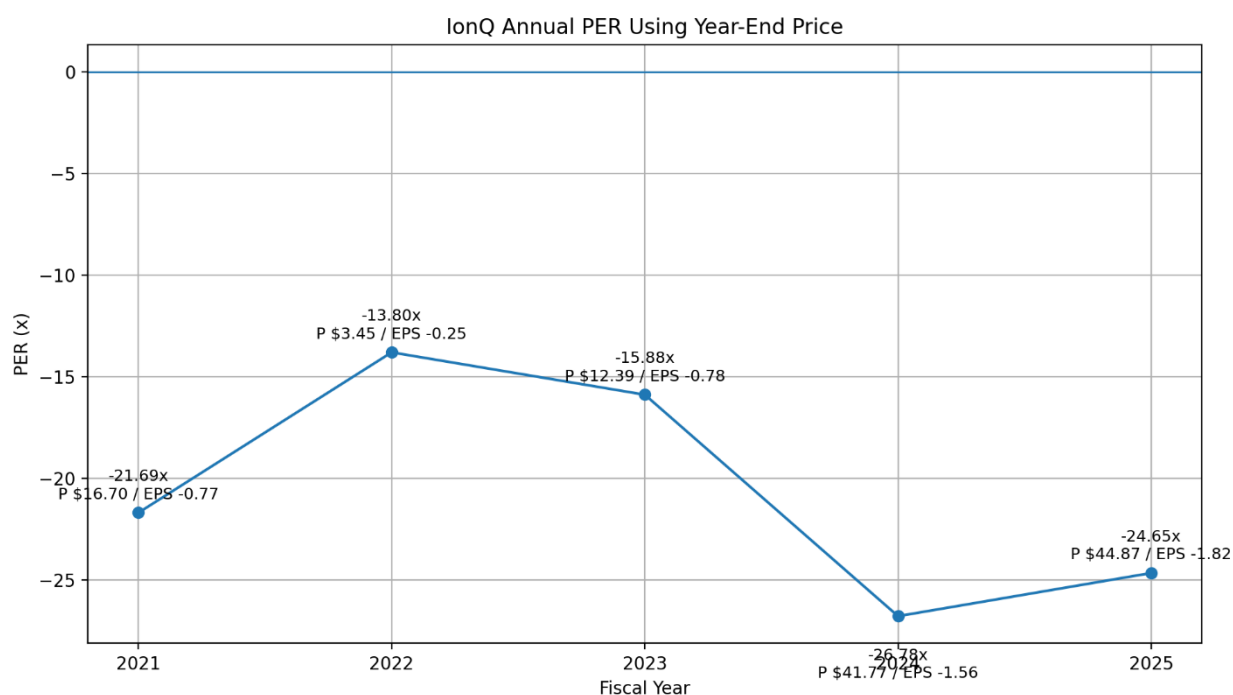
[Investor Ratio]

IonQ's ownership structure consists of approximately **54.71% institutional** investors, **0.55% executives** and directors, and **44.74% individual** and other investors. In other words, institutional ownership exceeds half of the total, while management ownership remains very low.

4-2. Financial Statement Health

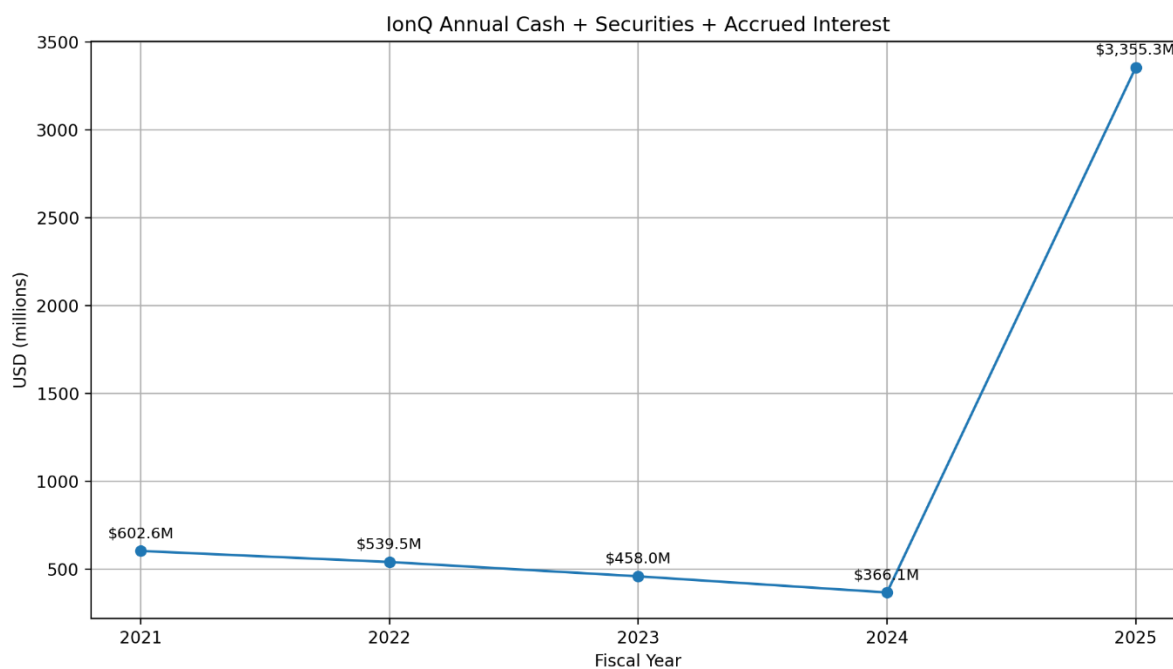
As of the end of March 2026, the company held approximately \$3.1 billion in cash, cash equivalents, and investments, indicating very strong short-term liquidity and debt repayment capacity. However, operating cash outflow reached approximately \$151 million in the first quarter of 2026, and operating losses continued, meaning long-term financial health depends on the rate of cash burn and the extent of any additional shareholder dilution.

5. Key Financial Metrics



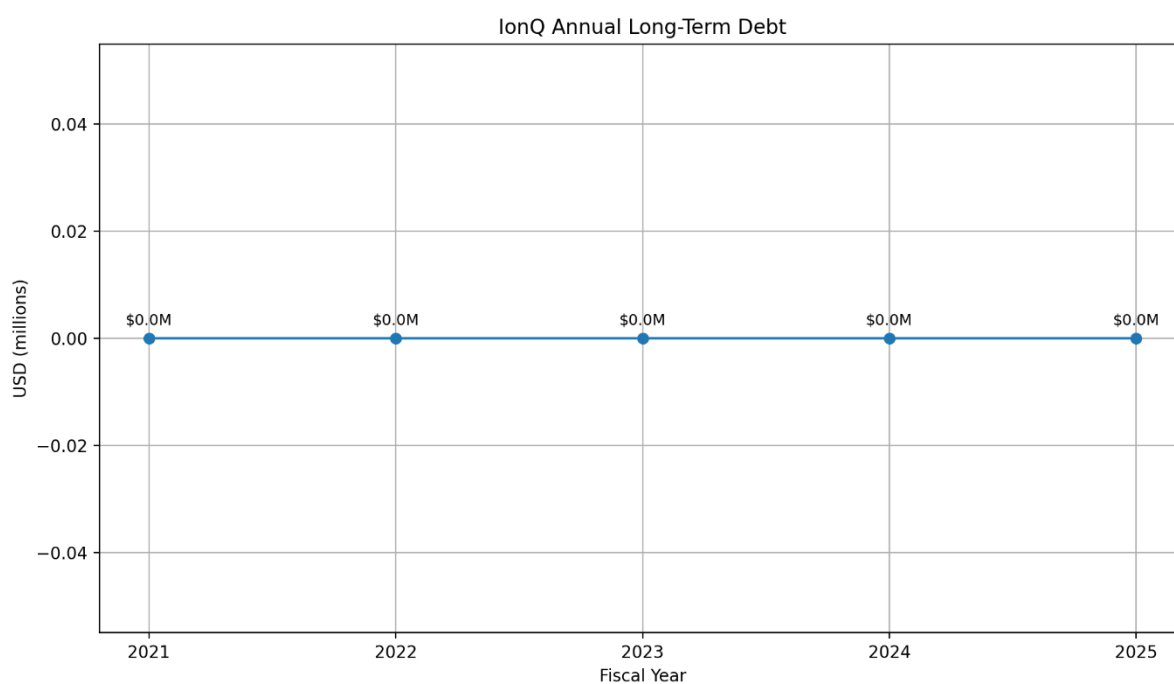
Formula: fiscal year-end closing price / annual diluted EPS. Negative PER is mathematically calculated but is not conventionally meaningful for valuation.

[Price Earnings Ratio]



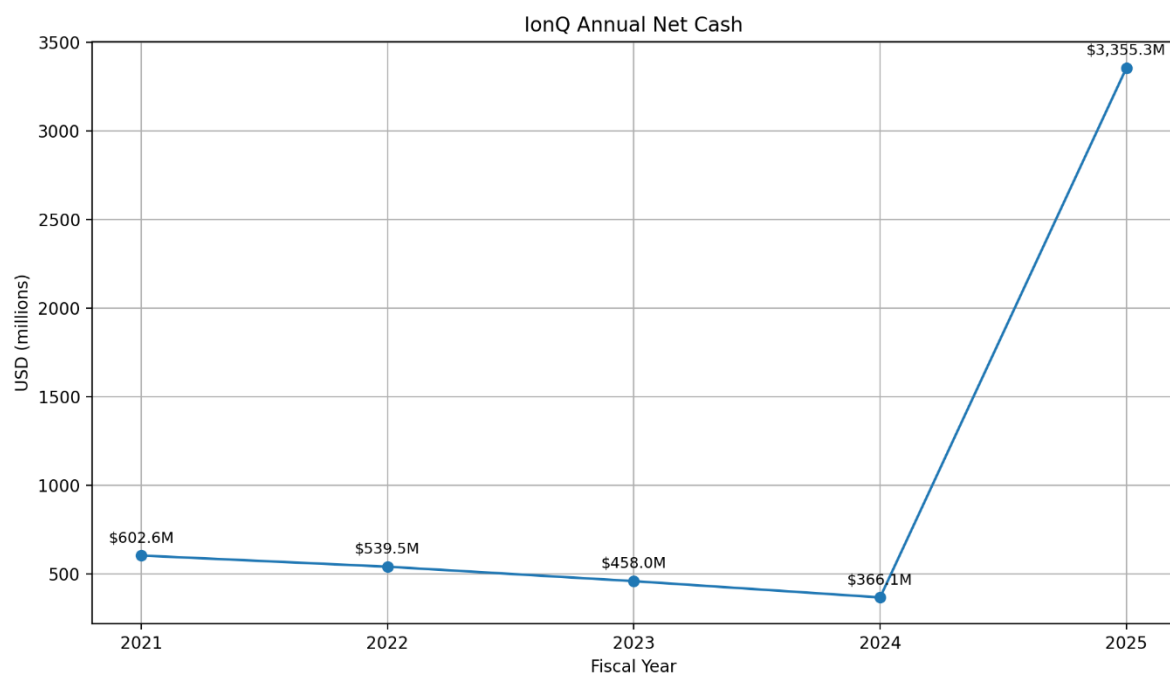
2021 value excludes accrued interest receivable because it was not separately disclosed in the FY2021 annual report.

[Total Cash and Cash Equivalents]



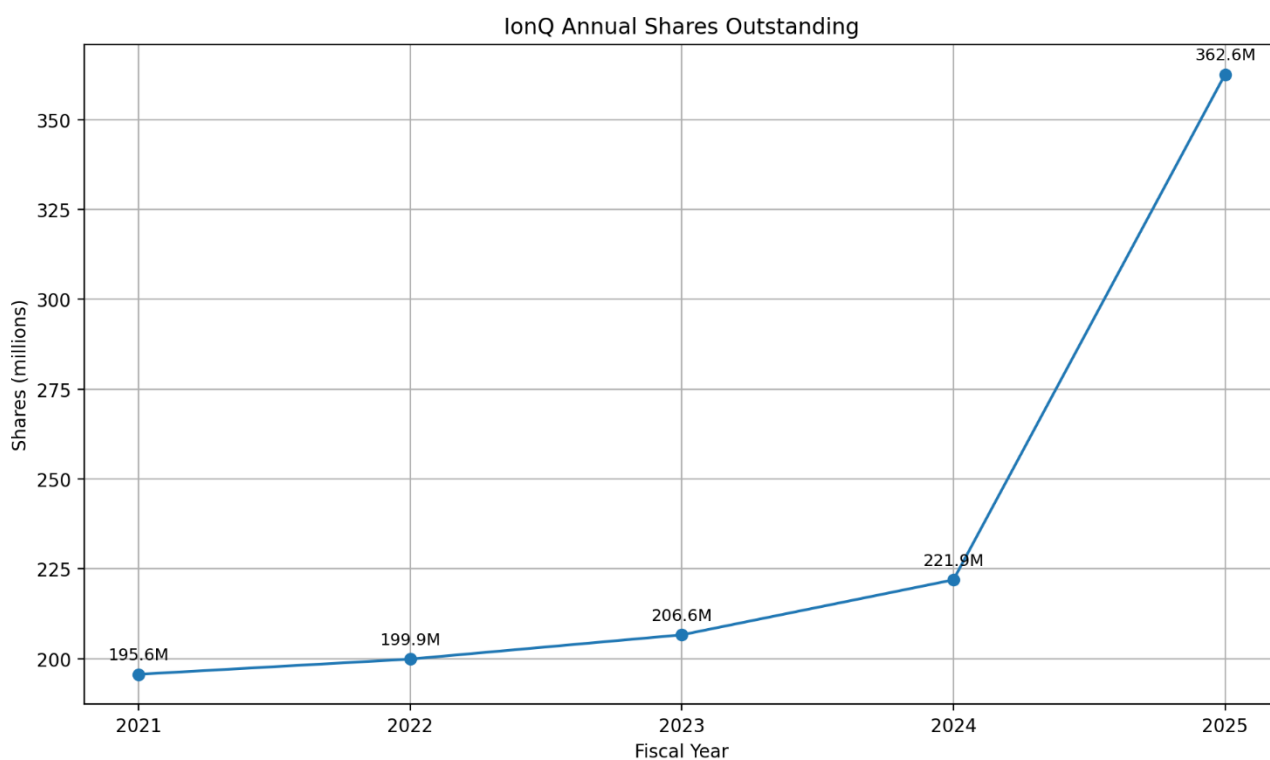
Operating lease liabilities are excluded from this chart.

[Long-Term Debt]

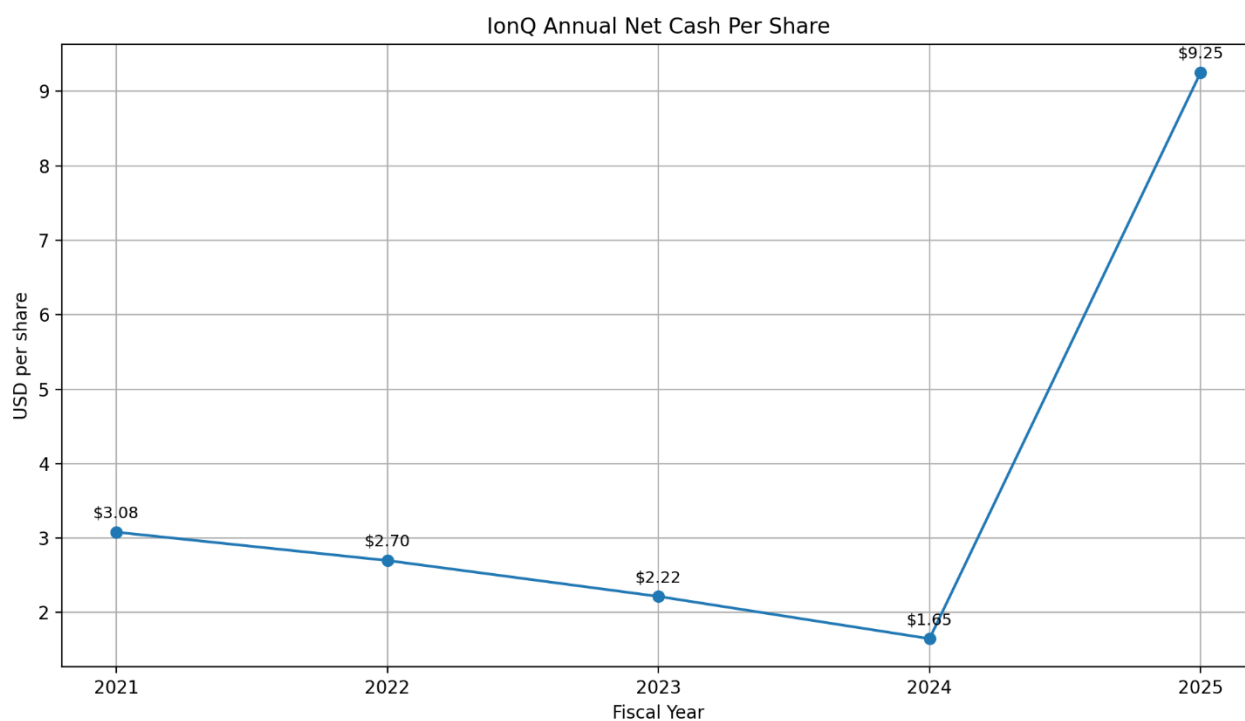


Net cash = Cash + Securities + Accrued Interest - Long-Term Debt. 2021 accrued interest was not separately disclosed.

[Net Cash]

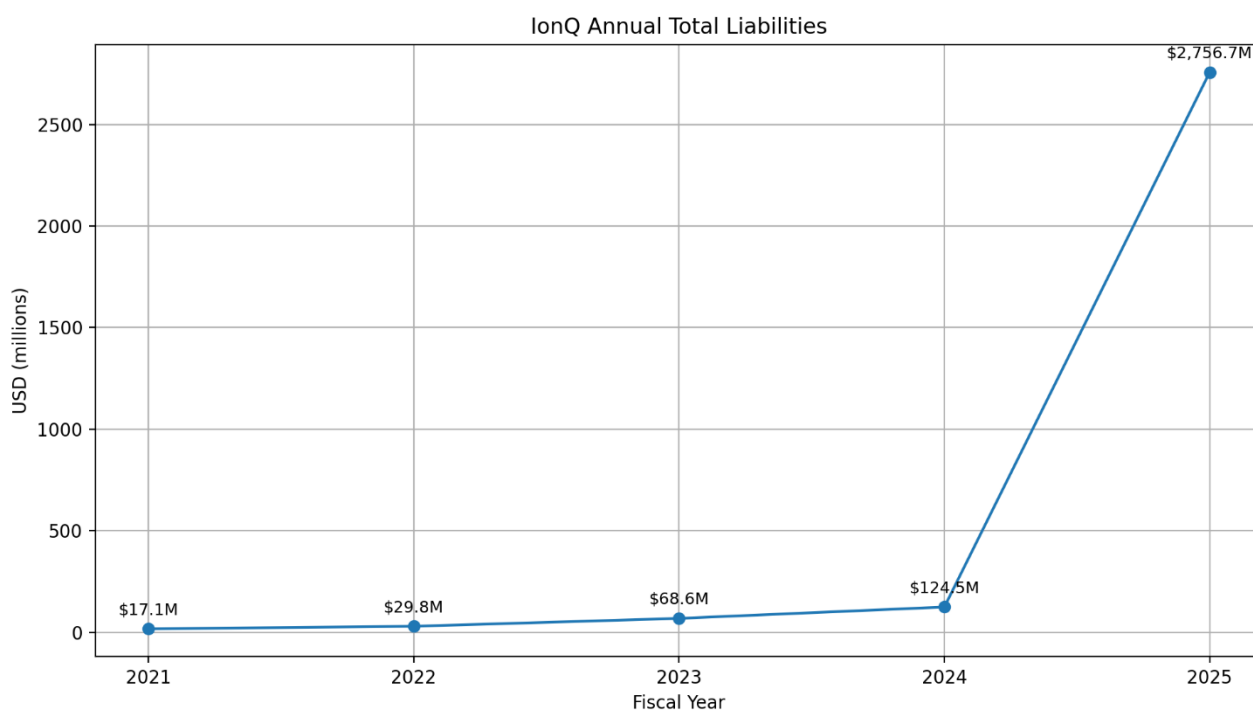


[Total Shares Outstanding]



2021 value is a lower-bound estimate because FY2021 accrued interest receivable was not separately disclosed.

[Net Cash per Share]



Total liabilities include warrant liabilities, lease liabilities, accounts payable, accrued liabilities, and unearned revenue.

[Total Liabilities]

Cash and cash equivalents increased sharply to approximately \$3.36 billion in 2025, but this was not generated through operations; it resulted from financing through large-scale stock and prepaid warrant* issuances. Total liabilities also increased to approximately \$2.76 billion, but most of this consisted not of conventional borrowings, but of warrant liabilities measured based on the share price. Therefore, cash flow risk and equity dilution risk should be assessed separately.

*Warrant: A derivative security that grants the holder the right to buy or sell a specific stock or asset at a predetermined price, known as the exercise price, on or before the expiration date.

6. Final Investment Analysis

6-1. Strengths

1) Large Cash Reserves

Cash and cash equivalents increased to approximately \$3.36 billion by the end of 2025, providing the company with substantial financial flexibility to support long-term investments in research and development, mergers and acquisitions, and manufacturing capacity expansion.

2) Virtually No Interest-Bearing Debt Burden

The company has no short-term or long-term interest-bearing debt, limiting cash flow pressure from rising interest rates or principal and interest repayments. Most of its total liabilities also consist of warrant valuation liabilities rather than conventional borrowings.

3) Expansion of Technology and Business Portfolio

Through acquisitions such as Oxford Ionics, the company is expanding into a comprehensive quantum technology company by securing capabilities not only in quantum computing, but also in networking, cybersecurity, sensing, and semiconductor manufacturing.

6-2. Weakness

1) Persistent Operating Losses and Uncertain Profitability

Research and development expenses and operating costs remain high relative to revenue, and the company has yet to generate stable operating profits. Therefore, its future cash burn rate and commercialization performance must be closely monitored.

2) Risk of Shareholder Dilution

The number of shares outstanding increased significantly in 2025 due to large-scale issuances of new shares and prepaid warrants, and the per-share ownership value of existing shareholders may be further diluted if additional warrants are exercised in the future.

6-3. Key Summary

Since its public listing in 2021, IonQ has consistently recorded net losses and a negative P/E ratio, while maintaining a capital structure with virtually no interest-bearing short-term or long-term debt.

In 2025, the company took advantage of its elevated share price to raise approximately \$3.0 billion through issuances of common stock and prepaid warrants, increasing its cash and cash equivalents to approximately \$3.36 billion. At the same time, total liabilities surged to approximately \$2.76 billion, but most of these liabilities consisted not of bank borrowings, but of warrant liabilities measured based on fluctuations in the share price.

The funds raised are being used for quantum computing research and development, acquisitions such as Oxford Ionics, expansion into quantum networking, cybersecurity, and sensing, and the internalization of manufacturing capabilities. However, the number of shares outstanding increased significantly in 2025, and additional warrant exercises remain possible, making shareholder dilution a key risk for existing investors.

Therefore, IonQ is currently

**“Company with strong cash liquidity that must be evaluated alongside
its continuing losses and cash burn, acquisition performance, and dilution burden.”**